



MINING DEALER BEST PRACTICE

Multi-Purpose Tool for Assembly Pump GP Water Auxiliary

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

This best practice was developed to reduce human involvement during the pump GP auxiliary assembly by using a unique tool that Ferreyros developed for this purpose. This best practice gives the specification and design requirements to build the tool, as well as the process steps to implement and use the tool in the assembly process.

2.0 Best Practice Description

The tool might be used with 3508 692, 3512 662, 3516 722, 3508 746, 3508B 26R 3512 8BR, 3512 962, 3516 2PK, 3516B 77R, 785C APX 992G 994 94F 777D 3PR, 785D DMC engines. This tool will be used for installing the bearing P/N 127-5400, ceramic seal, impeller and cover following a sequence below.

The design specs for the tool are shown in the supporting attachments section, and below is what the finished tool will look like.

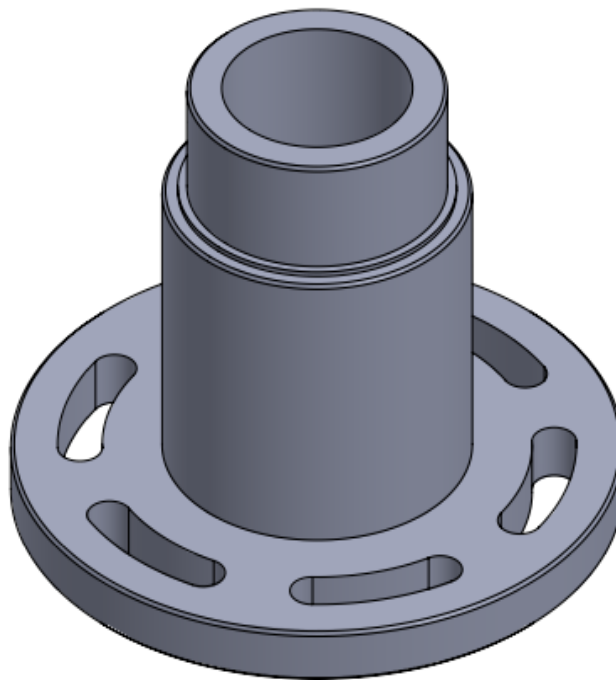


Figure 1: Multi-purpose tool.

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3.0 Implementation Steps

1. Install the seal on the pump GP auxiliary.

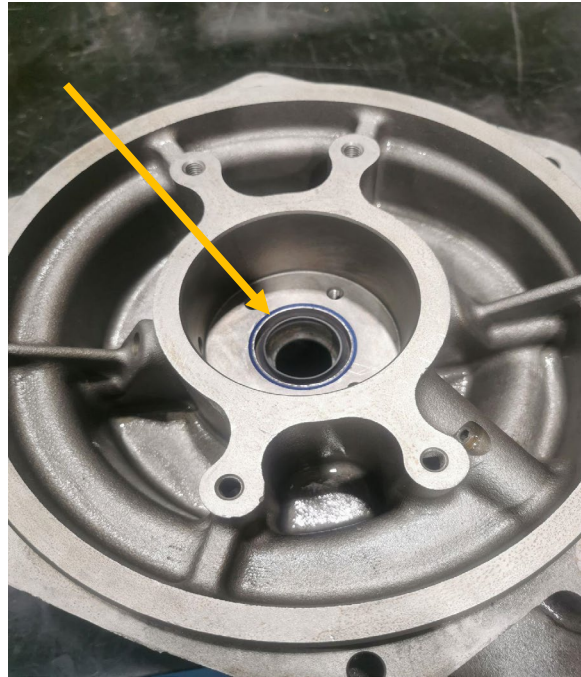


Figure 2: Seal installation.

2. Install the bearing P/N 127-5400 using the multi-purpose tool on the hydraulic press as shown in the figure below.



Figure 3: Bearing installation.

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3. Rotate the auxiliary GP pump and install the ceramic seal. Use the multi-purpose tool like figure 4 shows.



Figure 4: Ceramic seal installation using the multi-purpose tool

4. Place the auxiliary GP pump on the multi-purpose tool. Use the tool to avoid axial displacement during the impeller installation.



Figure 5: Impeller installation using multi-purpose tool.

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5. Install the pump cover using the multi-purpose tool.

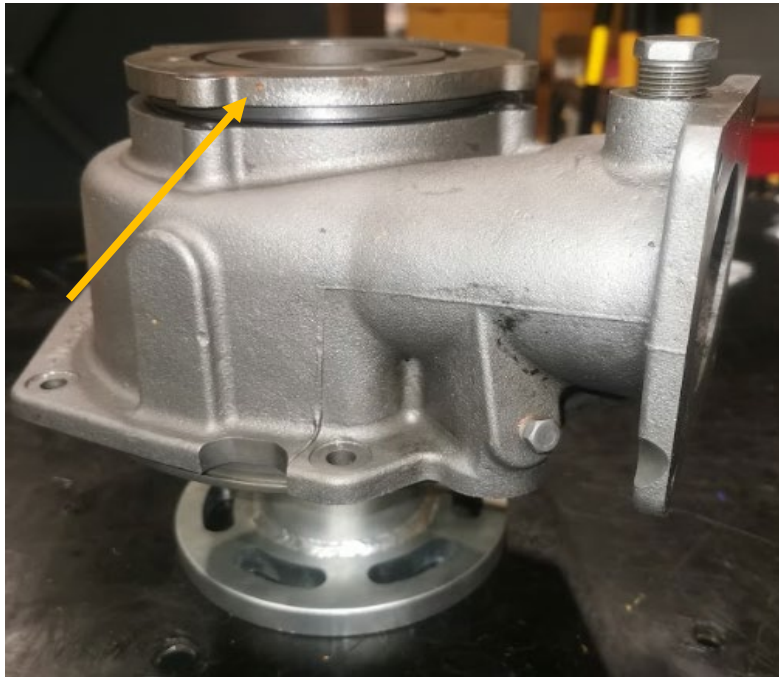


Figure 6: Impeller cover with the multi-purpose tool.

4.0 Benefits

- Decreases human effort during the installation of the bearing, seal, impeller and cover during pump GP auxiliary assembly;
- Simplifies the assembly process for the pump GP water auxiliary.

5.0 Resources Required

Item	Cost
Multi-Purpose Tool	\$650

Tool was made by a third party.

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8.0 Acknowledgements

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